

Topic- IoT-Based Smart Door Security System With Remote Face Enrollment and Real-Time Mobile Authorization.

1. Introduction

In today's world, security is a major concern for homes, offices, and organizations. Traditional lock-and-key systems are not secure enough and can be easily misused or lost. To overcome these limitations, smart security systems using IoT (Internet of Things) and mobile applications are becoming popular.

This project presents an IoT-based smart door security system that uses face recognition and mobile authorization to allow only authorized persons to enter. The system also sends real-time alerts to the owner if an unknown person is detected.

2. Objective

The main objectives of this project are:

- To develop a smart door system using IoT technology
- To allow access only to authorized persons
- To provide remote access control using a mobile application
- To send real-time alerts when unauthorized access is detected
- To improve security using face recognition and mobile verification

3. System Overview

This system works using a camera module (ESP32-CAM) that captures the image of a person standing near the door. The system checks whether the person is authorized or not.

- If the person is authorized, the door opens automatically
- If the person is unauthorized, their image is sent to the owner's mobile app
- The owner can approve or reject access remotely

4. Components Required

Hardware Components

- ESP32-CAM (Camera module with WiFi)

- Raspberry Pi 4 (optional for advanced processing)
- Relay module
- Solenoid door lock
- Power supply
- Buzzer (optional)

Software Components

- Android Studio (for app development)
- Firebase (for database and notifications)
- Face Recognition (OpenCV / AI model)
- Embedded C / Arduino IDE (for ESP32 programming)

5. System Architecture

The system consists of three main parts:

1. Image Capture Unit (ESP32-CAM)
2. Processing Unit (ESP32 / Raspberry Pi / Cloud)
3. User Interface (Android App)

Flow:

- Camera captures face
- Face is processed and compared
- Decision is taken (Authorized / Unauthorized)
- Notification sent to mobile
- Owner gives permission

6. Working Principle

Step 1: Face Detection

The ESP32-CAM continuously monitors the entrance and detects a face when someone arrives.

Step 2: Face Recognition

The captured image is compared with stored images in the database.

Step 3: Decision Making

Case 1: Authorized Person

- Face matches stored data

- Door unlocks automatically

Case 2: Unauthorized Person

- Face not recognized
- Image captured and sent to mobile app
- Notification sent to owner

Step 4: Mobile Authorization

- Owner receives alert with image
- Owner can:
 - Approve → Door opens
 - Reject → Door remains locked

7. Android Application Features

- User login and authentication
- Add/remove authorized users
- Upload face images
- Receive real-time notifications
- View captured images
- Approve or deny access
- Maintain access logs

8. Technologies Used

- Internet of Things (IoT)
- Face Recognition (AI / OpenCV)
- Firebase Cloud Database
- Firebase Cloud Messaging (FCM)
- Android Development
- WiFi Communication

9. Advantages

- High level of security
- Remote access control

- Real-time notifications
- No need for physical keys
- Easy to manage users

10. Disadvantages

- Requires internet connection
- Face recognition may fail in low light
- Initial setup cost
- Privacy concerns

11. Applications

- Smart homes
- Offices
- Laboratories
- Restricted areas
- Hostels and apartments

12. Future Enhancements

- Add fingerprint sensor
- Improve AI accuracy
- Add voice recognition
- Cloud storage for data
- Integration with smart home systems

13. Conclusion

The IoT-based smart door security system provides a modern and efficient way to secure access to rooms and buildings. By combining face recognition and mobile authorization, the system ensures that only authorized persons can enter. It also provides real-time monitoring and remote control, making it highly useful for modern security applications.

14. References

- IoT concepts and applications
- ESP32-CAM documentation
- Firebase official documentation
- Android Studio developer guide
- OpenCV face recognition tutorials
- Research papers on smart security systems
- Embedded systems textbooks
- Wireless communication basics
- AI and machine learning resources
- Smart home automation studies